
PHYSICS

9702/43

Paper 4 A Level Structured Questions

May/June 2016

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the May/June 2016 series for most Cambridge IGCSE®, Cambridge International A and AS Level components and some Cambridge O Level components.



Page 2	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

- 1 (a) (gravitational) potential at infinity defined as/is zero B1
- (gravitational) force attractive so work got out/done as object moves from infinity
(so potential is negative) B1 [2]
- (b) (i) $\Delta E = m\Delta\phi$
 $= 180 \times (14 - 10) \times 10^8$ C1
 $= 7.2 \times 10^{10} \text{ J}$ A1
increase B1 [3]
- (ii) energy required $= 180 \times (10 - 4.4) \times 10^8$
or
energy per unit mass $= (10 - 4.4) \times 10^8$ C1
 $\frac{1}{2} \times 180 \times v^2 = 180 \times (10 - 4.4) \times 10^8$
or
 $\frac{1}{2} \times v^2 = (10 - 4.4) \times 10^8$ C1
 $v = 3.3 \times 10^4 \text{ ms}^{-1}$ A1 [3]
- 2 (a) e.g. time of collisions negligible compared to time between collisions
no intermolecular forces (except during collisions)
random motion (of molecules)
large numbers of molecules
(total) volume of molecules negligible compared to volume of containing vessel
or
average/mean separation large compared with size of molecules
any two B2 [2]
- 2 (b) (i) mass $= 4.0 / (6.02 \times 10^{23}) = 6.6 \times 10^{-24} \text{ g}$
or
mass $= 4.0 \times 1.66 \times 10^{-27} \times 10^3 = 6.6 \times 10^{-24} \text{ g}$ B1 [1]
- (ii) $\frac{3}{2} kT = \frac{1}{2} m \langle c^2 \rangle$ C1
 $\frac{3}{2} \times 1.38 \times 10^{-23} \times 300 = \frac{1}{2} \times 6.6 \times 10^{-27} \times \langle c^2 \rangle$
 $\langle c^2 \rangle = 1.88 \times 10^6 \text{ (m}^2\text{s}^{-2}\text{)}$ C1
r.m.s. speed $= 1.4 \times 10^3 \text{ ms}^{-1}$ A1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

- 3 (a) acceleration/force proportional to displacement (from fixed point) M1
acceleration/force and displacement in opposite directions A1 [2]
- (b) maximum displacements/accelerations are different B1
graph is curved/not a straight line B1 [2]
- (c) (i) $\omega = 2\pi / T$ and $T = 0.8 \text{ s}$ C1
 $\omega = 7.9 \text{ rad s}^{-1}$ A1 [2]
- (ii) $a = (-)\omega^2 x$ C1
 $= 7.85^2 \times 1.5 \times 10^{-2}$
 $= 0.93 \text{ ms}^{-2}$ or 0.94 ms^{-2} A1 [2]
- (iii) $\Delta E = \frac{1}{2} m \omega^2 (x_0^2 - x^2)$ C1
 $= \frac{1}{2} \times 120 \times 10^{-3} \times 7.85^2 \times \{(1.5 \times 10^{-2})^2 - (0.9 \times 10^{-2})^2\}$ C1
 $= 5.3 \times 10^{-4} \text{ J}$ A1 [3]
- 4 (a) (i) product of speed and density M1
reference to speed in medium (and density of medium) A1 [2]
- (ii) α : ratio of reflected intensity and/to incident intensity B1
 Z_1 and Z_2 : (specific) acoustic impedances of media (on each side of boundary) B1 [2]
- (b) in muscle: $I_M = I_0 e^{-\mu x}$
 $= I_0 \exp(-23 \times 3.4 \times 10^{-2})$ C1
 $I_M / I_0 = 0.457$ C1
at boundary: $\alpha = (6.3 - 1.7)^2 / (6.3 + 1.7)^2$
 $= 0.33$ C1
 $I_T / I_M = [(1 - \alpha)] = 0.67$ C1
 $I_T / I_0 = 0.457 \times 0.67$
 $= 0.31$ A1 [5]

Page 4	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

5 (a) (i) 1011 A1 [1]

(ii)

0	0.25	0.50	0.75	1.00	1.25	1.50
1011	0110	1000	1110	0101	0011	0001

All 6 correct, 2 marks. 5 correct, 1 mark.

A2 [2]

(b) sketch: 6 horizontal steps of width 0.25 ms shown

M1

steps at correct heights and all steps shown

A1

steps shown in correct time intervals

A1 [3]

(c) increase sampling frequency/rate

M1

so that step width/depth is reduced

A1

increase number of bits (in each number)

M1

so that step height is reduced

A1 [4]

6 (a) sketch: from $x = 0$ to $x = R$, potential is constant at V_s

B1

smooth curve through (R, V_s) and $(2R, 0.5V_s)$

B1

smooth curve continues to $(3R, 0.33V_s)$

B1 [3]

(b) sketch: from $x = 0$ to $x = R$, field strength is zero

B1

smooth curve through (R, E) and $(2R, 0.25E)$

B1

smooth curve continues to $(3R, 0.11E)$

B1 [3]

7 (a) line has non-zero intercept/line does not pass through origin

B1

charge is/should be proportional to potential (difference)

or

charge is/should be zero when p.d. is zero

(therefore there is a systematic error)

B1 [2]

Page 5	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

- (b) reasonable attempt at line of best fit B1
- use of gradient of line of best fit clear M1
- $C = 2800 \mu\text{F}$ (allow $\pm 200 \mu\text{F}$) A1 [3]
- (c) energy = $\frac{1}{2} CV^2$ or energy = $\frac{1}{2} QV$ and $C = Q / V$ C1
- $\Delta \text{ energy} = \frac{1}{2} \times 2800 \times 10^{-6} \times (9.0^2 - 6.0^2)$ C1
- $= 6.3 \times 10^{-2} \text{ J}$ A1 [3]
- 8 (a) op-amp has infinite/(very) large gain B1
- op-amp saturates if $V^+ \neq V^-$ M1
- V^+ is at earth potential so P (or V^-) must be at earth A1 [3]
- (b) input resistance to op-amp is very large
or
current in R_2 = current in R_1 B1
- $V_{\text{IN}} (-0) = IR_2$ and $(0) - V_{\text{OUT}} = IR_1$ M1
- $V_{\text{OUT}} / V_{\text{IN}} = -R_1 / R_2$ A1 [3]
- (c) relay coil connected between V_{OUT} and earth M1
- correct diode symbol connected between V_{OUT} and coil or between coil and earth M1
- correct polarity for diode ('clockwise') A1 [3]
- 9 (a) 0.10 mm B1 [1]
- (b) $V_H = (0.13 \times 3.8) / (6.0 \times 10^{28} \times 0.10 \times 10^{-3} \times 1.60 \times 10^{-19})$ C1
- $= 5.1 \times 10^{-7} \text{ V}$ A1 [2]
- 10 (a) (non-uniform) magnetic flux in core is changing M1
- induces (different) e.m.f. in (different parts of) the core A1
- (eddy) currents form in the core M1
- which give rise to heating A1 [4]

Page 6	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

- (b) as magnet falls, tube cuts magnetic flux M1
e.m.f./ (eddy) currents induced in metal/aluminium (tube) A1
(eddy) current heating of tube M1
with energy taken from falling magnet A1
or
(eddy) currents produce magnetic field (M1)
that opposes motion of magnet (A1)
so magnet B has acceleration $< g$
or
magnet B has smaller acceleration/reaches terminal speed A1 [5]
- 11 (a) period = 15 ms C1
frequency $(= 1 / T) = 67 \text{ Hz}$ A1 [2]
- (b) zero A1 [1]
- (c) $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$ C1
 $= 0.53 \text{ A}$ A1 [2]
- (d) energy $= I_{\text{r.m.s.}}^2 \times R \times t$ or $\frac{1}{2} I_0^2 \times R \times t$
or
power $= I_{\text{r.m.s.}}^2 \times R$ and energy = power $\times t$ C1
energy $= 0.53^2 \times 450 \times 30 \times 10^{-3}$
 $= 3.8 \text{ J}$ A1 [2]
- 12 (a) (in a solid electrons in) neighbouring atoms are close together (and influence/interact with each other) M1
this changes their electron energy levels M1
(many atoms in lattice) cause a spread of energy levels into a band A1 [3]

Page 7	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – May/June 2016	9702	43

- (b) photons of light give energy to electrons in valence band B1
 electrons move into the conduction band M1
 leaving holes in the valence band A1
 these electrons and holes are charge carriers B1
 increased number/increased current, hence reduced resistance B1 [5]

13 (a) e.g. background count (rate)/radiation

multiple possible counts from each decay

radiation emitted in all directions

dead-time of counter

(daughter) product unstable/also emits radiation

self-absorption of radiation in sample or absorption in air/detector window

three sensible suggestions, 1 each B3 [3]

(b) $A = A_0 \exp(-\ln 2 \times t / T_{1/2})$

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\ln 2 \times 42.0 / T_{1/2})$$

or

$$1.21 \times 10^2 = 3.62 \times 10^4 \exp(-\lambda \times 42.0)$$

C1

$$T_{1/2} = 5.1 \text{ minutes (306 s)}$$

A1 [2]

(c) discrete energy levels (in nuclei)

B1 [1]